SALTDRAG · FEASIBILITY ASSESSMENT

SaltDrag feasibility assessment

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Sets per pan decided (D25), 2026-10-25
0.1	2026-10-24	Amish Chadha	First feasibility assessment (kit 1.8 upgrade), on the design after the STEM check fixes

SaltDrag feasibility assessment

SaltDrag is feasible to build and, on paper, to work: the physics sits within known science (SDG-STM-001), every load-bearing part checked keeps a margin of 2 or more at twice the rated pull after nine detail fixes (SDG-CAL-001; SDG-DEC-001 D17 to D24), and a town workshop can make it from plate, tube and bought parts for about USD 2,897; what is not yet known is whether it gathers salt faster per worker than the dantala and whether the forest department will allow it, and neither can be settled on paper. Method: STANDARDS section 23.

Table 1. Feasibility by aspect.

Aspect	Assessment	Rating	Risks and notes	Evidence
Design	A geared drum winch, a rope over a roller and a skid-mounted rake: each is long-established practice. Crank effort 89 N, cranking power up to 71 W, rope pull 3.7 kN or more on a full drum, lowest strength margin 2.08 at 6 kN, tipping margin 1.56 (1.46 at 6 kN)	High	The drag per metre of rake (0.9 kN/m) is assumed; if a real bed drags more, the skids are raised or the rake shortened	SDG-STM-001 S1 to S17; SDG-CAL-001 Sections A to C; the drum barrel's rope pressure factor is open (S7)
Materials	Galvanised steel on the dry ground, UHMW-PE, FRP and 316 stainless in the pan, polyester rope; galvanising outlasts a season by a wide margin	Medium	316 stainless in near-saturated warm brine can suffer crevice corrosion (S19); duplex bolts are the fallback	SDG-STM-001 S18 to S21; BOM
Manufacturing and assembly	Cut, drilled and welded plate and hollow section; lathe work on the drum shaft and keyways; bought gears, bearings, anchors and rope; every bolted joint checked for assembly at D3	High	Gear cutting and galvanising are bought in; the cross-tube, guard and axle details were redrawn so they can be assembled (SDG-DEC-001 D19)	SDG-BLD-001; `cad/fidelity.json`; SDG-STM-001 S26
Power and controls	Two people cranking; no power source, no controls beyond the pawl and the band brake	High	Heat limits safe cranking to the cooler hours (S16)	SDG-STM-001 S15, S16
Supply of parts	Steel from stockists in Gandhidham or Ahmedabad; module 4 gears, flange bearings, helical anchors and polyester rope are catalogue items in India	Medium	Stainless-insert thermoplastic bearings and certified rope may need to come from a large city	BOM; SDG-MKT-001
Site and permission	The capstans need firm, dry ground behind the bunds; use inside the sanctuary needs the forest department's agreement	Low	Access conditions bar heavy machinery; whether a hand capstan is acceptable is unknown (SDG-REG-001 G2); soil strength for the anchors is assumed (S12)	SDG-REG-001; SDG-STM-001 S12
Usefulness	About 532 kg/h with a crew of four, 133 kg per worker-hour; one set gathers about a third of a family pan's yearly output in a 61-day harvest	Medium	The dantala baseline is unmeasured (PR-005); one set per pan may not be enough for a family; it stays the design case until the partner answers (SDG-DEC-001 D25, decided 2026-10-25)	SDG-CAL-001 [D8], [D9], [D11]; SDG-OPT-001

Ratings: High, Medium, Low.